

MSC QUANTUM SCIENCE AND TECHNOLOGY

MASTER THESIS

Toward ML-Based State Discrimination on FPGA in Open-Source Quantum Control Electronics

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Abstract

Low-latency and high-fidelity qubit state measurement is essential for superconducting quantum computing, particularly for enabling mid-circuit measurements and real-time feedback. While commercial control hardware increasingly supports on-FPGA state discrimination, these platforms operate on closed firmware that precludes custom signal processing pipelines. Open-source FPGA frameworks offer an alternative path: full programmatic access to the readout chain, enabling researchers to integrate custom algorithms — including machine learning — directly into the measurement workflow.

This thesis documents the setup and characterisation of a QICK-based [1] readout system on a Xilinx ZCU216 RFSoc platform, covering FPGA configuration, remote operation, DDS-based waveform generation, and analogue front-end characterisation. The work establishes a functional signal chain capable of driving and reading out superconducting resonators, and examines the hardware constraints relevant to dispersive qubit readout, including balun response and filter rolloff across the relevant frequency bands.

Building on this infrastructure, we design and evaluate a machine-learning-assisted state discrimination pipeline using existing experimental single-shot readout data from a superconducting qubit device. We find that on integrated IQ data, linear discriminant analysis is already near-optimal — no MLP classifier improves upon it — consistent with the Gaussian blob structure of the IQ clouds. The only meaningful ML payoff comes from a lightweight 1D CNN trained on physically-grounded synthetic raw ADC traces, which recovers shots corrupted by mid-readout T_1 relaxation that integrated methods cannot distinguish: the CNN achieves 99.9% overall fidelity and 100% accuracy on relaxation shots (58.2 percentage points above LDA), noting that the latter figure is based on approximately 91 relaxation events in the test set. This model is trained and evaluated entirely offline; real-time deployment within the QICK firmware was not achieved within the scope of this work. Instead, the thesis concludes by outlining the path toward such deployment: the data collection procedure required with the ZCU216 platform, the model adaptation and quantisation constraints imposed by the programmable logic fabric, and the experimental methodology needed to assess whether deployment yields measurable improvements in classification speed, fidelity, or both.

Keywords: superconducting qubits, readout, FPGA, QICK, RFSoc, dispersive measurement, mid-circuit measurement, machine learning, state discrimination

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1 Introduction

Quantum computing holds the promise of solving classically intractable problems in optimisation, simulation, and cryptography. Among the leading hardware platforms, superconducting qubits stand out for their relatively fast gate times, high coherence, and compatibility with existing microwave engineering infrastructure [2]. A central challenge that must be overcome on the road to fault-tolerant operation is *qubit state measurement*: reading out the quantum information stored in a qubit quickly, accurately, and with minimal back-action on unmeasured qubits.

1.1 Motivation

Classical error correction requires measurements that are not only high-fidelity but also *fast* relative to qubit coherence times, and — crucially for quantum error correction — it requires *mid-circuit measurements*: the ability to measure ancilla qubits repeatedly during a quantum algorithm, use the results to extract error syndromes in real time, and continue the computation on the data qubits. This capability demands a control electronics stack capable of making a state discrimination decision on-chip in microseconds — the timescale of the readout integration window itself — rather than the milliseconds typical of a software round-trip to a host CPU.

Field-Programmable Gate Arrays (FPGAs) offer a compelling solution. Their deterministic, sub-microsecond timing enables real-time signal processing close to the qubit hardware. Commercial control platforms such as Qblox, Zurich Instruments, and Quantum Machines already exploit this, providing on-FPGA state discrimination within closed, proprietary firmware stacks. However, closed firmware precludes custom signal processing pipelines: researchers cannot inject bespoke algorithms — including machine-learning classifiers — into the real-time decision path. The Quantum Instrumentation Control Kit (QICK) [1] addresses this gap as an open-source FPGA firmware and software framework that turns Xilinx Radio-Frequency System-on-Chip (RFSoc) boards into complete qubit controllers, combining fast digital-to-analogue converters (DACs), analogue-to-digital converters (ADCs), and a tightly integrated soft-processor into a single compact platform. Full access to the programmable logic fabric makes it possible to define the entire readout and discrimination pipeline, including the integration of custom ML models, in a way that closed commercial firmware does not permit.

1.2 Goals and Contributions

Despite the availability of QICK, assembling a *general-purpose* readout pipeline from an uncharacterised board to a working single-shot state discriminator requires integrating many layers: physical wiring, FPGA configuration, phase calibration, signal optimisation, and machine-learning post-processing. The lack of a unified reference implementation makes this process time-consuming and error-prone, especially for multi-qubit scenarios.

This thesis makes the following contributions:

1. **Full-stack integration** of the QICK framework on a Xilinx ZCU216 RFSoc board with an XM655 analogue front-end module, including remote operation via a persistent Pyro4 server.
2. **Phase-coherent readout calibration**: a systematic procedure for measuring and compensating the random DAC–ADC phase offset that arises at every board power-on, enabling reproducible IQ measurements across sessions.

3. **Readout data-path design:** a characterisation of the trade-offs between decimated and accumulated readout modes and a complete resonator spectroscopy workflow suitable for both single-qubit and multiplexed measurements.
4. **Real-time feedback primitives:** demonstration of the `condj` conditional-branch primitive on the tProcessor in loopback, and design of an active qubit reset sequence combining dispersive readout, thresholding, and a calibrated π pulse — establishing the infrastructure needed for mid-circuit measurement once the board is connected to a qubit chip.
5. **ML-assisted state discrimination:** offline design and evaluation of a lightweight ML discrimination pipeline on existing experimental SSRO data, establishing that LDA is near-optimal for integrated IQ inputs, and demonstrating via synthetic raw traces that a 1D CNN recovers T_1 -relaxation shots irrecoverable by integration-based methods. A concrete path toward FPGA deployment is outlined.

1.3 Thesis Outline

Section 2 reviews the relevant theory of superconducting qubits, circuit quantum electrodynamics (cQED), and dispersive readout. Section 3 describes the hardware platform. Section 4 covers system integration and calibration. Section 5 presents the readout architecture and measurement workflows. Section 6 describes real-time feedback and mid-circuit measurement. Section 7 evaluates machine-learning-based state discrimination. Section 8 summarises the results and outlines future directions.

2 Theoretical Background

2.1 Superconducting Qubits

Superconducting qubits are anharmonic LC oscillators whose nonlinearity is provided by one or more Josephson junctions [2]. The Josephson junction acts as a nonlinear inductor with energy $E_J \cos \hat{\varphi}$, where $\hat{\varphi}$ is the superconducting phase across the junction and E_J is the Josephson energy. Combined with a shunt capacitance C (charging energy $E_C = e^2/2C$), the system forms an artificial atom with a discrete, anharmonic spectrum that can be addressed at the transition frequency ω_{01} . Several qubit designs have been developed that exploit this anharmonicity in different parameter regimes; this thesis is primarily concerned with two: the transmon and the fluxonium.

Transmon. The transmon [3] operates deep in the regime $E_J \gg E_C$, making its energy levels insensitive to charge noise. Its transition frequency lies typically in the 4 GHz to 8 GHz range, and its anharmonicity $\alpha = \omega_{12} - \omega_{01}$ is of order $-E_C/\hbar$, typically a few hundred MHz.

Fluxonium. The fluxonium [4] replaces the transmon’s large capacitor with a superinductance L (inductive energy $E_L = (\Phi_0/2\pi)^2/L$, where Φ_0 is the flux quantum), threading the loop with an external flux Φ_z . The resulting double-well potential can be tuned between a plasmon regime (near $\Phi_z = 0$) and a fluxon regime (near $\Phi_z = \Phi_0/2$). Fluxonium qubits offer long coherence times and are of particular interest because their readout landscape is richer than the transmon’s: the dispersive shift χ varies strongly with flux bias, and near the half-flux sweet spot the two eigenstates carry opposite persistent currents, giving rise to a large state-dependent dispersive shift that can be exploited for high-contrast readout [4].

2.2 Circuit Quantum Electrodynamics

In the circuit QED (cQED) architecture [5], a qubit is coupled dispersively to a microwave resonator. The resonator acts both as a readout bus and as a filter that protects the qubit from radiative decay into the measurement chain. The qubit–resonator Hamiltonian in the dipole-coupling (Rabi) form (setting $\hbar = 1$) is

$$\hat{H} = \hat{H}_{\text{qubit}} + \omega_r \hat{a}^\dagger \hat{a} + g \hat{n}_q (\hat{a} + \hat{a}^\dagger), \quad (1)$$

where ω_r is the resonator frequency, $\hat{a}$ ($\hat{a}^\dagger$) is the resonator annihilation (creation) operator, g is the qubit–resonator coupling strength, and $\hat{n}_q$ is the relevant qubit operator (charge for capacitive coupling) [2]. When the qubit–resonator detuning is large compared to the coupling ($|g| \ll |\omega_{ij} - \omega_r|$), second-order perturbation theory in g yields an effective qubit-state-dependent shift of the resonator frequency, derived in Section 2.3 below.

2.3 Dispersive Readout

When $|g_{ij}| \ll |\omega_{ij} - \omega_r|$ for all relevant qubit transitions $i \leftrightarrow j$, second-order perturbation theory in g yields a qubit-state-dependent shift of the resonator frequency [6]:

$$\hat{H}_{\text{disp}} = \left(\omega_r + \sum_i \chi_i |i\rangle \langle i| \right) \hat{a}^\dagger \hat{a}, \quad \chi_i = \sum_{j \neq i} \frac{|g_{ij}|^2}{\omega_{ij} - \omega_r} - \frac{|g_{ji}|^2}{\omega_{ji} - \omega_r}, \quad (2)$$

where $g_{ij} = \langle i | g \hat{n}_q | j \rangle$ and $\omega_{ij} = \omega_j - \omega_i$. The textbook two-level dispersive shift $\chi = \chi_1 - \chi_0$ follows from truncating to the computational subspace, but for fluxonium-class qubits the *straddling regime* (where the resonator frequency lies between the two lowest qubit transition frequencies, $\omega_{01} < \omega_r < \omega_{12}$) makes this truncation unsafe: the dispersive shifts from the $0 \rightarrow 1$ and $1 \rightarrow 2$ transitions carry opposite signs and partially cancel, so higher levels must be included [3].

In practice, one probes the resonator at its bare frequency ω_r and observes a state-dependent phase shift of the transmitted or reflected signal. For a single-qubit system, the resonator sits at $\omega_r + \chi_0$ when the qubit is in $|0\rangle$ and at $\omega_r + \chi_1$ when in $|1\rangle$, resulting in an IQ-plane separation that grows with the dimensionless cavity pull $2|\chi|/\kappa$, where κ is the resonator linewidth [5, 7].

Fluxonium dispersive readout. For a fluxonium qubit, the dispersive shift landscape is richer than for the transmon because the qubit spectrum varies strongly with the external flux Φ_z . Near the half-flux point $\Phi_z = \Phi_0/2$, the double-well potential becomes symmetric and the two lowest eigenstates carry opposite persistent currents. The resulting state-dependent coupling to the resonator produces a dispersive shift $\chi = \chi_1 - \chi_0$ whose magnitude depends strongly on the flux bias point, making the readout contrast highly tunable. This flux dependence is a key motivation for the fluxonium readout optimisation work being pursued in related projects within the Qilimanjaro team.

2.4 Direct Digital Synthesis

Direct Digital Synthesis (DDS) generates analogue waveforms by accumulating a phase register at a fixed clock rate and mapping the instantaneous phase value to a digital amplitude via a look-up table. A DDS numerically controlled oscillator (NCO) produces a tone at frequency

$$f_{\text{out}} = \frac{f_{\text{clk}} \cdot \Delta\phi}{2\pi}, \quad (3)$$

where f_{clk} is the system clock, $\Delta\phi$ is the phase increment per clock cycle, and N is the bit width of the phase accumulator. DDS avoids the spur and image problems associated with analogue IQ mixers because all imperfections are digital and predictable. In QICK, the tProcessor drives DDS engines on both the DAC (transmit) and ADC (receive) sides, enabling phase-locked signal generation and coherent demodulation [1].

2.5 State Discrimination

Following dispersive readout, the measured IQ quadrature amplitudes (I, Q) form two clusters in the complex plane — one for $|0\rangle$ and one for $|1\rangle$. Single-shot state discrimination assigns each measurement to one of these clusters. The simplest approach is a linear threshold: a hyperplane in the IQ plane that minimises the total misclassification probability, which for Gaussian clusters of equal covariance is the perpendicular bisector of the line connecting the centroids [8]. The single-shot readout fidelity is

$$\mathcal{F} = 1 - \frac{1}{2} (p(1|0) + p(0|1)), \quad (4)$$

where $p(1|0)$ is the probability of assigning $|1\rangle$ given that the qubit was prepared in $|0\rangle$, and vice versa. More sophisticated classifiers — quadratic discriminant analysis, neural networks — can improve fidelity when the clusters are non-Gaussian or when multi-level leakage is present.

2.6 Machine Learning for State Discrimination

The state discrimination problem defined in Section 2.5 is a binary classification task: given a measurement outcome $\mathbf{x} \in \mathbb{R}^d$, assign it to one of two classes ($|0\rangle$ or $|1\rangle$). The choice of classifier and its optimality depend on the statistical structure of the data.

Linear Discriminant Analysis. If the two classes are distributed as multivariate Gaussians with equal covariance, $p(\mathbf{x} | s) = \mathcal{N}(\boldsymbol{\mu}_s, \boldsymbol{\Sigma})$ for $s \in \{0, 1\}$, then the Bayes-optimal decision boundary is the hyperplane

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1} \left(\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_0 + \boldsymbol{\mu}_1) \right) = 0, \quad (5)$$

which is exactly the decision boundary found by Linear Discriminant Analysis (LDA) [8]. Because this boundary is already optimal under the Gaussian equal-covariance assumption, no more expressive classifier — including multilayer perceptrons (MLPs) — can improve on it. An MLP can at best approximate this boundary; it cannot exceed the Bayes optimum. When the covariances differ between classes, Quadratic Discriminant Analysis (QDA) uses class-specific covariances $\boldsymbol{\Sigma}_0, \boldsymbol{\Sigma}_1$ and yields a curved boundary that is Bayes-optimal under the unequal-covariance Gaussian model.

Limitations of integrated IQ input. Both LDA and QDA operate on a single integrated (I, Q) pair per shot — the time-averaged quadrature amplitudes over the readout window. This compression discards all temporal information. In particular, if the qubit relaxes from $|1\rangle$ to $|0\rangle$ at some time t^* within the readout window, the integrator averages over both signal levels and produces a point that falls between the two clusters. No classifier operating on the integrated input can recover these shots, regardless of its expressive power, because the information about the true prepared state has been destroyed by the integration.

Convolutional Neural Networks on raw traces. A one-dimensional convolutional neural network (1D CNN) takes the full raw ADC time trace as input, preserving the temporal structure that integration discards. A CNN applies a bank of learned filters across the time axis via the convolution operation

$$(f * x)[t] = \sum_{\tau=0}^{k-1} f[\tau] x[t - \tau], \quad (6)$$

where f is a learned filter of length k and x is the input signal. By stacking convolutional layers, the network can detect local temporal features at increasing scales — in the readout context, the step discontinuity in the IQ trajectory that occurs when the qubit relaxes mid-shot. This makes CNNs well-suited to recovering T_1 -relaxation shots that are irrecoverable from integrated IQ, at the cost of requiring raw trace data rather than accumulated IQ pairs.

3 Hardware Platform

3.1 Xilinx ZCU216 RFSoc

The Xilinx Zynq UltraScale+ ZCU216 evaluation board [9] is built around an RFSoc device that integrates, on a single chip, a quad-core ARM Cortex-A53 processor, an FPGA fabric, and 16 RF-DAC and 16 RF-ADC channels with sampling rates up to 9.85 GS/s and 2.5 GS/s, respectively. Operating at 12-bit and 14-bit resolution, the converters can directly synthesise and capture signals in the 0 GHz to 6 GHz range without an external frequency up-conversion stage, provided the analogue signal chain is suitably conditioned.

The board runs the PYNQ Linux distribution, which exposes the FPGA fabric and the converters through Python libraries, making it straightforward to implement and iterate on firmware/software stacks. Table 1 summarises the key specifications relevant to this thesis.

Table 1: Selected ZCU216 specifications relevant to qubit readout.

Parameter	Value	Note
DAC channels	16	Differential GSPS
DAC rate	up to 9.85 GS/s	Per channel
DAC resolution	14 bit	
ADC channels	16	Differential GSPS
ADC rate	up to 2.5 GS/s	Per channel
ADC resolution	12 bit	
FPGA fabric	UltraScale+	930k LUT
Processor	4× Cortex-A53	1.5 GHz
Operating system	PYNQ Linux	

3.2 QICK Software and Firmware

The Quantum Instrumentation Control Kit (QICK) [1, 10] is an open-source framework originally developed at Fermilab that turns an RFSoc board into a self-contained qubit controller. It consists of two layers:

Firmware. A set of Vivado IP blocks synthesised and placed onto the FPGA fabric, comprising:

- **Signal generators:** DDS-based waveform engines that drive the DACs with arbitrary envelopes, constant tones, or flat-top pulses. Generator 6 (physical port 2_231 on the ZCU216) is used as the primary readout generator throughout this work.
- **Readout blocks:** ADC-side DDS demodulators that down-convert the received signal to baseband. Each readout block provides both a decimated output (time-domain I/Q waveform) and an accumulation buffer (averaged I/Q pair). Readout channel 0 (ADC port 0_226) corresponds to the first ADC pair on the XM655 breakout.
- **tProcessor V2:** a soft-processor that orchestrates pulse timing with nanosecond precision. It executes a custom instruction set including direct memory writes (`memri`), register arithmetic (`mathi`, `regwi`), conditional branches (`condj`), and pulse triggers. The ZCU216 uses the 64-bit variant of the tProcessor, which extends the original 32-bit design with wider arithmetic and richer branching.

Software. A Python library (the `qick` package) that runs on the ARM core and exposes high-level classes for writing and running experiments:

- `QickSoc`: instantiates the full hardware stack, loads the bitstream, and provides access to the converters and tProcessor.
- `QickProgram` / `AveragerProgram` / `RAveragerProgram`: abstractions for writing measurement sequences at increasing levels of automation. `RAveragerProgram` enables FPGA-internal parameter sweeps via the tProcessor’s `update()` hook, avoiding Python overhead between experiment points.
- Remote operation: a persistent Pyro4 server running on the board makes it possible to control the board from a laptop over a standard network connection, without reinitialising the FPGA between notebook sessions. This design also preserves the phase calibration across reconnects.

3.3 XM655 Analogue Front-End Module

The ZCU216’s RF converters use differential signalling on High-Density (HD) headers (JHC3 and JHC5 on the board). The XM655 module is Xilinx’s daughter-card solution: it provides BALUN (balanced-to-unbalanced) transformers that convert the differential DAC outputs and ADC inputs to/from single-ended SMA ports, along with optional filtering.

A key constraints introduced by the XM655 in this work:

- The BALUNs on the XM655 cover the frequency range 10 MHz to 6000 MHz. Resonator frequencies above 6 GHz require connections directly on the HD header pins using pigtail cables.

The loopback path used for calibration experiments consists of: DAC port 2_231 (JHC3) → HD2-SMA pigtail → BALUN → SMA cable → BALUN → HD2-SMA pigtail → ADC port 0_226 (JHC5).

Figure 1 shows the high-level signal flow from the tProcessor to the qubit chip and back.

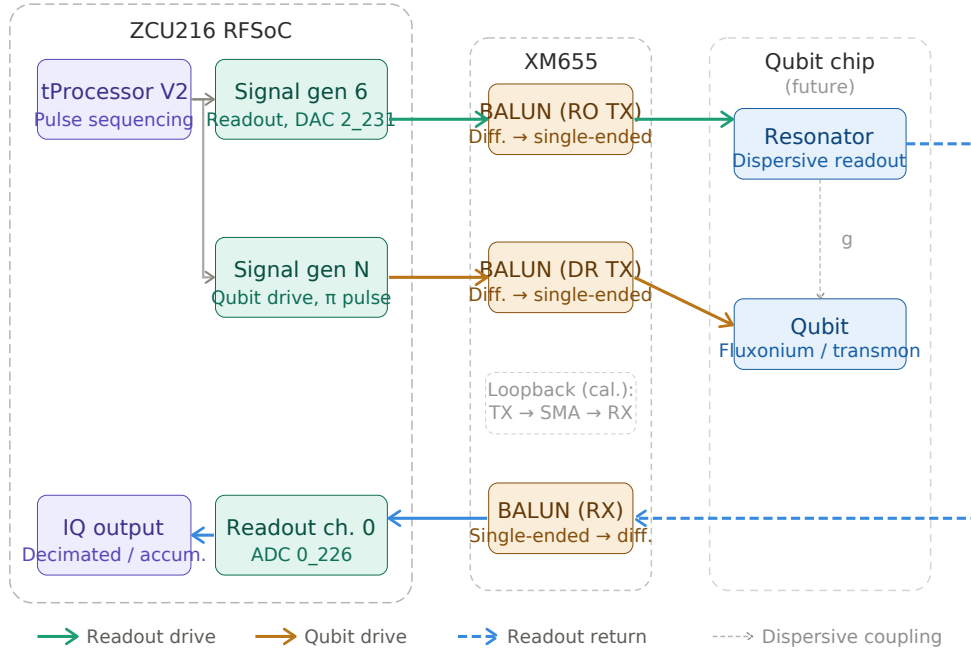


Figure 1: System signal chain for the QICK-based readout platform. The tProcessor V2 sequences pulses on two generator channels: signal gen 6 (DAC 2_231) drives the readout tone through the XM655 TX BALUN to the resonator; signal gen N drives the qubit directly via a second TX BALUN channel for π -pulse control (the specific channel number depends on the final wiring and is left as N here). The reflected readout tone returns through the XM655 RX BALUN to ADC 0_226 and is demodulated to an IQ pair by readout channel 0. The XM655 BALUNs cover 10 MHz to 6000 MHz; resonator frequencies above 6 GHz require direct HD-header pigtail connections, bypassing the XM655. The qubit chip and its connections are shown as a design target; the board was not connected to a qubit chip within the scope of this work. Calibration experiments use the loopback path (TX BALUN $\rightarrow$ SMA cable $\rightarrow$ RX BALUN) shown inside the XM655 container.

4 System Integration

4.1 FPGA Configuration and Board Bring-up

The ZCU216 boots the PYNQ Linux image from an SD card. After boot, the QICK bitstream is loaded onto the FPGA fabric by instantiating a `QickSoc` object in Python, which calls the PYNQ overlay loader and configures all IP blocks. Because the RFSoc converters are controlled through AXI register maps rather than external chip-select lines, initialisation is entirely software-driven.

The board is accessed over SSH. To avoid re-initialising the hardware for every new notebook session, a persistent Pyro4 object server is launched once on the board and kept running:

```

1 # On the board (runs once per power cycle)
2 import Pyro4, qick
3 soc = qick.QickSoc()
4 daemon = Pyro4.Daemon()
5 uri = daemon.register(soc, "qick.soc")
6 Pyro4.locateNS().register("qick.soc", uri)
7 daemon.requestLoop()

```

From a remote laptop:

```

1 import Pyro4
2 soc = Pyro4.Proxy("PYRONAME:qick.soc")
3 soccfg = soc # lightweight config object

```

[TODO: Consider removing as I never actually used this, but it is possible]

This design preserves the phase calibration state (see Section 4.3) across reconnects and makes experiment iteration fast.

4.2 Channel Mapping

The QICK firmware assigns each physical DAC output to a numbered *generator channel* and each physical ADC input to a numbered *readout channel*. On the ZCU216 with the default QICK bitstream, the mapping relevant to this work is shown in Table 2.

Table 2: QICK channel mapping on the ZCU216.

QICK Channel	Physical Port	HD Header	Function
Generator 6	DAC 2_231	JHC3	Readout / control drive
Readout 0	ADC 0_226	JHC5	IQ acquisition

4.3 Phase Calibration

Each time the ZCU216 is powered on or the FPGA bitstream is reloaded, the DDS oscillators in the DAC and ADC blocks boot with a random mutual phase offset ϕ . Without compensation, IQ measurements rotate unpredictably between sessions, making it impossible to compare results across power cycles.

Phase model. A loopback tone from DAC 2_231 to ADC 0_226 measures ϕ at a given frequency. Because the two oscillators are effectively separated by a fixed time delay τ (the total group delay around the loopback path), the phase offset scales linearly with frequency:

$$\phi(f) = \phi_0 - 360^\circ \cdot \tau \cdot f, \quad (7)$$

where ϕ_0 is a constant offset and τ is extracted from the fit. Note that the fitted τ reflects the combined pipeline delay of the DAC and ADC signal chains — including the digital up- and down-conversion stages (DUC/DDC) and internal FIFOs — rather than the physical cable propagation delay, which for a bench loopback of 1–2 m contributes only a few nanoseconds.

Calibration procedure. The phase calibration sweeps a range of frequencies f_i near the target readout frequency, records the averaged IQ phasor at each point, and extracts $\phi_i = \arg\langle I + jQ \rangle$. The pair (ϕ_0, τ) is then fitted using `scipy.optimize.least_squares`, taking care to handle the $0^\circ/360^\circ$ phase wrap correctly. The fit is considered reliable when the signal-to-noise ratio satisfies $|\text{mag}| \gg \text{RMS}$ (e.g. $637 \gg 1.8$ ADU in a typical calibration run).

The resulting `res_phase` value:

$$\phi_{\text{res}} = \phi_0 - 360^\circ \cdot \tau \cdot f_{\text{res}} \quad (8)$$

is stored in the experiment configuration dictionary and passed to the readout pulse via `soccfg.deg2reg(res_phase)`, which converts degrees to the integer register format expected by the tProcessor. This calibration must be re-run after every board reboot or bitstream reload, but remains valid throughout a measurement session.

Measurement cadence. A summary of the recommended start-of-session checklist is:

1. Power on and wait for PYNQ to boot (≈ 60 s).
2. Connect the loopback (DAC 2_231 $\rightarrow$ BALUN $\rightarrow$ SMA $\rightarrow$ BALUN $\rightarrow$ ADC 0_226) and run the phase calibration notebook.
3. Identify the correct `adc_trig_offset` using decimated readout (see Section 5.2).
4. Switch to accumulated readout for all data-taking.
5. At session end, call `soc.reset_gens()` to halt all running generators.

[TODO: Maybe add steps such as SSH, bring up Jupyter, etc.]

5 Readout Architecture

5.1 DDS-Based Signal Generation

In a standard heterodyne readout chain, a local oscillator (LO) at frequency f_{LO} is mixed with a baseband envelope to up-convert the readout tone to the resonator frequency f_r . This analogue mixing process introduces image products and LO leakage that degrade fidelity, requiring careful IQ imbalance correction.

QICK avoids analogue mixing entirely. Both the readout drive and the demodulation reference are produced by the on-chip DDS engines, which operate at the direct output frequency of the RFSoc converters (up to several GHz). The waveform generated for qubit readout has the form

$$s(t) = A \cdot \text{env}(t) \cdot \cos(2\pi f_r t + \phi_{\text{res}}), \quad (9)$$

where $\text{env}(t)$ is the pulse envelope (constant, Gaussian, DRAG, etc.), f_r is programmed directly into the DDS frequency register, and ϕ_{res} is the phase correction from calibration. The only noise sources are digital: quantisation, clock jitter, and DDS phase truncation spurs — all of which are deterministic and can be mitigated by appropriate filtering and sample-rate choice.

5.2 Decimated vs. Accumulated Readout

After the ADC samples the incoming signal, the QICK readout block demodulates it by multiplying with the NCO reference tone and low-pass filtering. Two output modes are available:

Decimated readout. Stores the full time-domain I/Q waveform over the readout window (e.g. 1000 samples). This is essential for setup: it reveals when the readout pulse arrives relative to the trigger (`adc_trig_offset`), whether the integration window is correctly positioned, and whether the signal shape is as expected. The single-shot noise is ~ 1.9 ADU RMS.

Accumulated readout. The FPGA sums all ADC samples within the readout window into a single (I, Q) pair in hardware — one pair per shot. This is the standard mode for SSRO: each shot yields one IQ point, and the discrimination classifier operates on that point. The per-shot integrated IQ noise is determined by the within-window averaging over T_{win} samples, not by the number of repeated shots. From the real SSRO data, the per-shot cluster widths are approximately 0.65–0.97 ADU per quadrature (state-dependent), with an IQ separation of 2.53 ADU and a projected SNR of 2.46, yielding a single-shot fidelity of 0.906 [11].

For applications that do *not* require single-shot resolution — such as resonator spectroscopy or Rabi calibration — the same shot can be repeated N_{reps} times and the results averaged, reducing the noise on the *mean* phasor as $1/\sqrt{N_{\text{reps}}}$:

$$\sigma_{\text{mean}} = \frac{\sigma_{\text{single}}}{\sqrt{N_{\text{reps}}}}. \quad (10)$$

This averaging is appropriate when only the average response is needed; it cannot be applied to SSRO since averaging destroys the per-shot state information.

The practical workflow is: (1) use `acquire_decimated()` to find the correct `adc_trig_offset` and verify signal quality; (2) switch to `acquire()` (accumulated) for all data-taking, using single shots for SSRO and repeat-averaged shots for spectroscopy and calibration experiments.

5.3 Resonator Spectroscopy

The resonator bare frequency ω_r and linewidth κ are found by sweeping the probe frequency around the estimated resonator frequency and measuring the transmitted or reflected IQ amplitude. In QUICK, there are two sweep strategies:

Fast sweep (RAveragerProgram). The tProcessor increments the DAC DDS frequency register at each experiment step via the `update()` method. This approach is fast because the FPGA handles all steps with no Python round-trips. However, the ADC NCO frequency is set via the ARM AXI bus and cannot be updated by the tProcessor. For resonator spectroscopy, both the DAC and ADC frequencies must track together; this is not achievable with the fast method.

Slow sweep (Python outer loop). A `for` loop in Python updates both the DAC generator frequency and the ADC readout frequency at each step. The per-step overhead is a few milliseconds (dominated by network round-trips in remote mode), but this approach can sweep *any* parameter, including the ADC NCO. This is the correct approach for resonator spectroscopy.

In practice:

```

1 freqs = np.linspace(f_start, f_stop, n_points)
2 amps = []
3 for f in freqs:
4     cfg["pulse_freq"] = f
5     prog = AveragerProgram(soccfg, cfg)
6     iq = prog.acquire(soc, load_pulses=True, progress=False)
7     amps.append(np.abs(iq[0][0][0] + 1j*iq[0][0][1]))

```

The resonator frequency is identified as the dip (reflection) or peak (transmission) in the swept amplitude response.

5.4 Multiplexed Readout

For multi-qubit systems, simultaneous readout of several qubits on the same physical line requires *multiplexed readout*: a broadband probe signal containing multiple tones at the individual resonator frequencies $\{f_{r,k}\}$ [11, 12]. The QICK framework supports this through a *Polyphase Filter Bank* (PFB) readout mode.

A PFB is a bank of digital bandpass filters implemented as a single, efficient DSP operation (polyphase decomposition of a prototype filter followed by an FFT). It takes the broadband ADC stream and simultaneously demodulates N equally-spaced frequency channels, analogous to a digital spectrum analyser. Advantages over the single-tone DDS approach include:

- No need to retune the ADC NCO between qubits.
- Computational cost scales as $O(N \log N)$ rather than $O(N)$ independent demodulators.
- Sidesteps the ADC-frequency-sweep limitation: each PFB output channel is a fixed digital filter, not a tunable DDS.

Multiplexed readout via the QICK PFB firmware was not implemented within the scope of this work. The single-tone DDS architecture described in Sections 5.2 and 5.3 was used throughout. A natural next step is to extend the architecture to cover the four Dátil resonator frequencies characterised at Qilimanjaro: 6.505, 6.730, 6.978, and 7.389 GHz. These frequencies lie above the 5 GHz to 6 GHz design passband of the XM655 BALUNs, so simultaneous multiplexed readout at all four tones would require direct HD-header connections with discrete BALUNs, as discussed in Section 3. The PFB channel spacing and prototype filter bandwidth would also need to be verified against the 184 MHz minimum separation between adjacent resonators in this device.

6 Real-Time Feedback and Mid-Circuit Measurement

A key motivation for FPGA-based control is the ability to execute *real-time* conditional logic: perform a measurement, make a decision, and apply a corrective pulse — all on the FPGA, without a round-trip to the host CPU. This section describes the tProcessor V2 instruction set, demonstrates the conditional branching primitive, and outlines the active qubit reset protocol.

6.1 tProcessor Architecture

The tProcessor V2 is a soft-processor synthesised in the FPGA fabric. It executes a custom instruction set tailored to pulse generation and readout sequencing. Key features include:

- **Register file:** 32 general-purpose 64-bit registers per channel page, plus cross-page registers for inter-channel synchronisation.
- **Pulse triggers:** dedicated instructions that fire DAC signal generators or arm ADC acquisition windows.
- **Timing instructions:** `synci` (advance tProcessor time without touching channel timelines), `sync_all` (synchronise all channels), `wait_all` (stall until all readout windows close).

- **Conditional branch:** `condj page, reg_a, op, reg_b, label` jumps to `label` if `reg_a op reg_b` is true.

All branching targets are resolved at *compile time* by the Python assembler: Python's sequential execution of `body()` builds an instruction list, `label()` records the current position as a named address, and the assembler backpatches jump targets before loading the program onto the tProcessor. By the time the program runs, only integer addresses exist; there is no runtime symbol table.

6.2 Conditional Logic Demonstration

The QUICK Demo 3 notebook demonstrates the core conditional-branch primitive:

```

1 def body(self):
2     cfg = self.cfg
3     self.trigger(adcs=[cfg["ro_ch"]],
4                 adc_trig_offset=cfg["adc_trig_offset"])
5     # condj: jump to SKIP_PULSE if number < threshold
6     self.condj(0, 2, '<', 1, 'SKIP_PULSE')
7     self.pulse(ch=cfg["res_ch"])          # skipped if condition true
8     self.label('SKIP_PULSE')
9     self.wait_all()
10    self.sync_all(self.us2cycles(cfg["relax_delay"]))

```

Registers 1 and 2 are pre-loaded with the threshold and the test value via `regwi`. When `number=100`, `threshold=50`: the condition `100 < 50` is false, the pulse fires, and the ADC captures a loopback signal. When `number=10`, `threshold=50`: the condition is true, the pulse is skipped, and the ADC trace is flat noise.

Timing subtlety. Inside a `body()` with an open ADC acquisition window, `synci` must be used to advance time between pulses, not `sync_all`. `sync_all` advances all channel timelines to the furthest point, which is the end of the ADC window; subsequent pulses would then be scheduled after the window has closed. `synci` advances only the tProcessor's internal counter, leaving the ADC timeline undisturbed.

6.3 π Pulse Calibration

A π pulse rotates the qubit state by 180° on the Bloch sphere: $|0\rangle \rightarrow |1\rangle$ and $|1\rangle \rightarrow |0\rangle$. It is a fundamental primitive for qubit initialisation (active reset) and single-qubit gates. Calibrating it requires a connected qubit chip; this section describes the procedure that would be followed once the ZCU216 is connected to a device.

The π pulse amplitude is found via a *Rabi experiment*: the qubit drive pulse amplitude is swept from zero, and the readout signal oscillates between the $|0\rangle$ and $|1\rangle$ IQ clusters as the qubit is driven through successive half-turns on the Bloch sphere. The first amplitude at which the signal crosses from the $|0\rangle$ cluster to the $|1\rangle$ cluster determines the π pulse amplitude. The Rabi sequence is:

1. Prepare qubit in $|0\rangle$ (wait $\gg T_1$).
2. Apply qubit drive pulse at frequency ω_{01} with variable amplitude A and fixed duration.
3. Read out via the resonator.

The qubit drive tone is at ω_{01} (the qubit transition frequency, typically 5 GHz to 7 GHz for fluxonium), which is distinct from the resonator frequency ω_r used for readout. Two separate generator channels are therefore required: one for readout, one for qubit control. The QICK firmware supports this natively — the tProcessor can address both channels independently with nanosecond timing precision.

6.4 Active Qubit Reset

Active reset rapidly returns the qubit to $|0\rangle$ by measuring its state and applying a corrective π pulse only if the qubit is found in $|1\rangle$ [13]. The QICK tProcessor V2 has all the primitives required to implement this entirely on-FPGA: the conditional branch demonstrated in Section 6.4 provides the decision logic, and the two-channel architecture supports simultaneous readout and drive. The following protocol was designed for deployment once the board is connected to a qubit chip:

1. **Dispersive readout:** fire the readout tone and acquire the accumulated IQ result.
2. **Thresholding:** the readout firmware compares the IQ result against a pre-calibrated threshold and writes a binary decision (0 or 1) into a tProcessor register.
3. **Conditional π pulse:** `condj` checks the decision register. If the qubit is in $|1\rangle$, a calibrated π pulse is applied on the qubit drive channel. Otherwise, the pulse is skipped.
4. **Wait:** allow any residual photons in the resonator to decay before starting the experiment.

Because the entire decision-to-pulse path would run inside the tProcessor, the latency between completing the readout window and firing the corrective pulse is determined solely by FPGA clock cycles. For the ZCU216 operating at a 384 MHz fabric clock, this latency is of order tens of nanoseconds — orders of magnitude faster than a software round-trip.

6.5 Latency Budget

Table 3 gives an estimated latency breakdown for the active reset sequence.

Table 3: Estimated latency contributions in the active reset loop.

Stage	Latency	Notes
Readout pulse duration	0.5 μ s to 2 μ s	Depends on κ
ADC integration window	0.5 μ s to 2 μ s	
FPGA decision (threshold)	~ 10 ns	tProcessor cycles
Conditional branch	~ 3 ns	1 tProc cycle
π pulse duration	50 ns to 500 ns	Depends on T_1
Total (excl. readout)	< 1 μ s	

The dominant latency is the readout integration window; the digital decision and conditional fire together contribute less than 100 ns, well within qubit coherence times of 10 μ s to 100 μ s for state-of-the-art fluxonium devices.

7 ML-Assisted State Discrimination

7.1 IQ Data and the Discrimination Problem

Each single-shot readout produces an averaged (I, Q) pair — a point in the complex plane. Repeated preparation and measurement of the qubit in the same state produces a cloud (cluster) of such points, whose centroid and spread depend on the readout power, resonator parameters, and integration time. Two such clusters, one for $|0\rangle$ and one for $|1\rangle$, must be distinguished reliably.

For ideal Gaussian clusters of equal covariance, the optimal linear classifier is the perpendicular bisector of the line segment joining the two centroids:

$$d(\mathbf{x}) = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1} \left(\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_0 + \boldsymbol{\mu}_1) \right) \underset{\hat{y}=0}{\overset{\hat{y}=1}{\geq}} 0, \quad (11)$$

where $\boldsymbol{\mu}_0, \boldsymbol{\mu}_1 \in \mathbb{R}^2$ are the class centroids and $\boldsymbol{\Sigma}$ is the common covariance matrix (Linear Discriminant Analysis, LDA [8]). When the covariances differ, Quadratic Discriminant Analysis (QDA) uses the class-specific covariances and yields a curved decision boundary.

In practice, residual state preparation errors, qubit T_1 decay during the readout pulse, and higher-level leakage ($|2\rangle, |3\rangle$) distort the clusters from ideal Gaussians. These effects motivate the use of more expressive classifiers. The assignment fidelity is defined as

$$\mathcal{F} = \frac{1}{2} [P(\hat{s} = 0 \mid s = 0) + P(\hat{s} = 1 \mid s = 1)], \quad (12)$$

the average per-class accuracy across both prepared states.

7.2 Lightweight Classifiers for FPGA Deployment

For eventual on-FPGA deployment, the classifier must be computable using only fixed-point arithmetic and a small number of multiply-accumulate operations. The key constraints are:

- **Latency:** the decision must complete within one tProcessor cycle budget ($\lesssim 10$ ns) to avoid extending the readout dead time.
- **Resource usage:** the classifier must fit in the available DSP slices and block RAM of the FPGA without displacing the readout and tProcessor logic.
- **Trainability:** the classifier must be trainable on a laptop from a few thousand calibration shots.

With these constraints in mind, three classifier families are considered. A **linear threshold** (LDA) reduces to a single dot product and a comparison — the minimum possible arithmetic, directly implementable in the tProcessor’s `mathi` instruction. A **quadratic threshold** (QDA) evaluates a 2×2 matrix product (4 multiply-accumulates) and handles asymmetric cluster covariances. A **small MLP** with inputs (I, Q) , one or two hidden layers of 4–64 neurons with ReLU activations, and a sigmoid output can be computed as a matrix-vector product and fits in $\lesssim 200$ 18-bit DSP slices on the ZCU216. The following sections report results from evaluating these classifiers offline on existing experimental data; FPGA deployment is discussed in Section 7.7. For prior work on neural-network-based discrimination, including multiplexed multi-qubit readout, see Lienhard et al. [14].

7.3 Experimental Data

The dataset used throughout this chapter is **SSR0.h5**, collected on a superconducting qubit device at Qilimanjaro. It contains 2000 single-shot IQ measurements per prepared state ($|0\rangle$ and $|1\rangle$), accumulated on-FPGA by the rectangular integrator in **axis_readout_v2**. Each shot is a single (I, Q) pair — the time-averaged quadrature amplitudes over the readout window. The class centroids and separations extracted from this dataset are summarised in Table 4.

Table 4: IQ blob statistics extracted from SSR0.h5.

Quantity	$ 0\rangle$	$ 1\rangle$
Centroid μ_I (ADU)	-0.32	-2.10
Centroid μ_Q (ADU)	$+4.50$	$+6.31$
IQ separation (ADU)	2.53	
IQ correlation	-0.587	-0.664

The negative IQ correlation in both states indicates a rotated noise ellipse, consistent with a readout tone that is not perfectly aligned to the I axis. The data were split 80/20 into training and test sets (3200 / 800 shots) using stratified sampling.

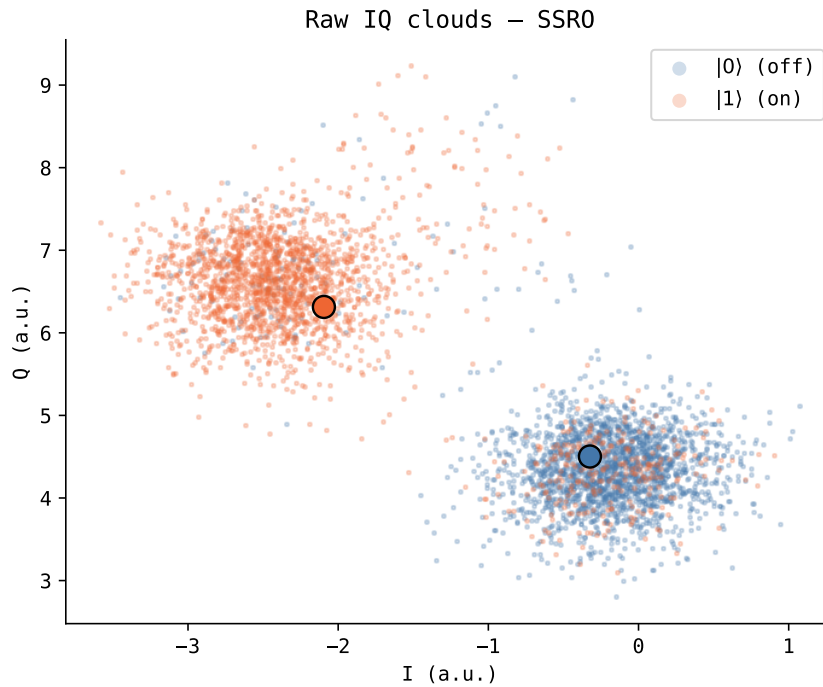


Figure 2: IQ scatter plot of the SSR0.h5 dataset (2000 shots per state). Blue: $|0\rangle$ (qubit off); orange: $|1\rangle$ (qubit on). Filled circles mark the cluster centroids. The two clouds are well-separated and approximately Gaussian, with the $|1\rangle$ ellipse visibly wider, consistent with the $\approx 1.7\times$ covariance asymmetry in Table 4. The diagonal orientation of both ellipses reflects the residual IQ rotation noted in the text.

7.4 Part I — Integrated IQ Discrimination

LDA, QDA, and a representative MLP (hidden layers $32 \rightarrow 16$, ReLU) were trained on the integrated IQ data. Table 5 reports the assignment fidelity on the held-out test set.

Table 5: Assignment fidelity on integrated IQ data (test set, 800 shots).

Classifier	$\mathcal{F}$	$ 0\rangle$ acc.	$ 1\rangle$ acc.
LDA	0.906	0.940	0.873
QDA	0.906	0.940	0.873
MLP (32 $\rightarrow$ 16)	0.906	0.940	0.873

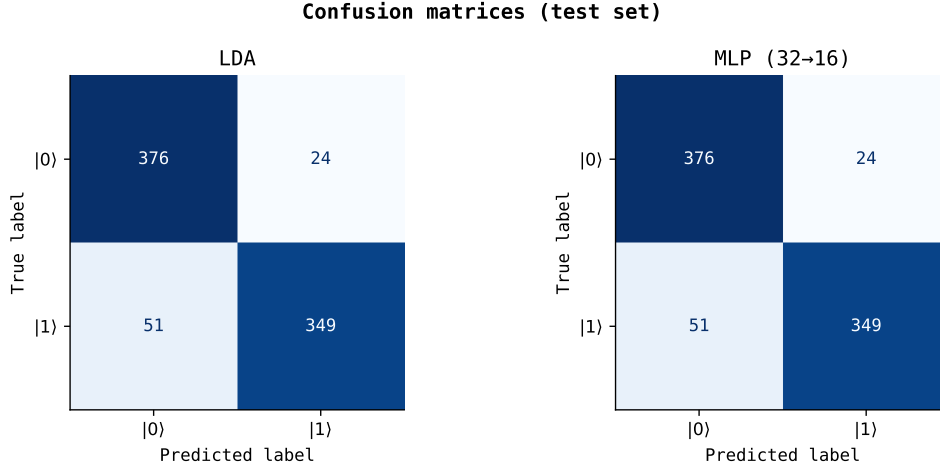


Figure 3: Confusion matrices for LDA and MLP (32 $\rightarrow$ 16) on the held-out test set (800 shots). Both classifiers produce identical assignments: 376/400 correct for $|0\rangle$ and 349/400 correct for $|1\rangle$. QDA produces the same matrix and is omitted. The asymmetry between off-diagonal entries (24 vs. 51) is the signature of T_1 relaxation biasing errors toward $|1\rangle \rightarrow |0\rangle$ misclassification.

All three classifiers achieve identical fidelity. The QDA result is particularly informative: the two IQ clouds do *not* share the same covariance — the $|1\rangle$ noise ellipse is approximately $1.7\times$ wider than the $|0\rangle$ ellipse (Table 4) — so QDA’s curved boundary is in principle more expressive than the linear LDA boundary. That QDA recovers no additional fidelity is explained by the large cluster separation: the Mahalanobis distance between the two centroids is approximately 2.4, placing the clusters well into the regime where the optimal boundary is nearly linear regardless of covariance asymmetry. LDA, QDA, and the MLP are all saturating the classification capacity of the two-dimensional integrated IQ input.

The $|0\rangle$ accuracy (94.0%) exceeds the $|1\rangle$ accuracy (87.3%), pointing at a systematic source of $|1\rangle$ misclassification. This asymmetry is the first diagnostic signature of T_1 relaxation during the readout window, which causes some $|1\rangle$ shots to produce an integrated IQ that falls in the $|0\rangle$ blob.

7.5 Part II — Architecture Search

To test whether the LDA–MLP parity in Part I is specific to the (32 $\rightarrow$ 16) architecture, a systematic grid search was performed over 20 MLP configurations spanning 1–3 hidden layers and 4–64 neurons per layer (17 to 6465 parameters), evaluated by 5-fold stratified cross-validation.

The result is striking: the fidelity landscape is entirely flat. No architecture — across three orders of magnitude in parameter count — improves on LDA. This is not a failure of the models; it is a statement about the data.

This result follows directly from the analysis in Part I. The two clusters are well-

Table 6: Architecture grid search results (5-fold CV fidelity). LDA baseline: 0.898 ± 0.013 . No MLP architecture exceeds LDA.

Architecture	$\mathcal{F}$ (CV)	$\pm$ std	Δ vs. LDA
(4,)	0.897	0.012	-0.001
(8,)	0.897	0.012	-0.001
(16,)	0.898	0.012	-0.000
(32,)	0.898	0.012	-0.000
(64,)	0.897	0.012	-0.001
(8, 8)	0.896	0.012	-0.002
(16, 16)	0.897	0.012	-0.001
(32, 32)	0.897	0.012	-0.001
(64, 64)	0.897	0.012	-0.001
(16, 8)	0.897	0.012	-0.001
(32, 16)	0.897	0.012	-0.001
(16, 16, 8)	0.898	0.012	-0.000
(32, 32, 16)	0.897	0.013	-0.001
(64, 64, 32)	0.897	0.012	-0.002

separated (Mahalanobis distance ≈ 2.4), placing them in the regime where the optimal decision boundary is nearly linear even accounting for the $\approx 1.7\times$ covariance asymmetry between states — as confirmed by QDA matching LDA exactly. LDA therefore finds a near-Bayes-optimal boundary on this input, and no classifier — regardless of expressive power — can improve on it using the same two-dimensional integrated IQ features.

The implication is fundamental: further fidelity improvement cannot come from a more expressive classifier on the same integrated IQ input. It must come from richer input features. The $|1\rangle$ accuracy deficit diagnosed in Part I identifies the mechanism directly: T_1 relaxation mid-shot.

7.6 Part III — Raw-Trace CNN

7.6.1 Motivation and the T_1 Relaxation Problem

As established in Section 2.6, any classifier operating on integrated IQ is fundamentally limited when T_1 relaxation occurs mid-shot: the integrator averages over both signal levels and produces a point that falls between the two clusters, and the information about the true prepared state has been destroyed. No classifier operating on this compressed input can recover these shots, regardless of its expressive power.

The raw ADC time trace, however, encodes the relaxation time t^* directly as a step discontinuity in the IQ trajectory. A 1D CNN trained on raw traces can detect this jump and correctly identify the prepared state even when the integrated IQ falls between the two clusters. Part III demonstrates this capability using physically-grounded synthetic traces, since raw decimated traces from the ZCU216/QICK setup are not yet available (acquiring them requires using `acquire_decimated` rather than the standard `acquire` call, and collecting a dedicated `SSRO_raw_synthetic.h5` dataset — the simulated counterpart used here — or real hardware traces with the board connected to a qubit chip). The theoretical motivation for using a CNN rather than a deeper MLP is discussed in Section 2.6: convolutional filters are well-suited to detecting local temporal structure — in this case the step discontinuity caused by mid-readout relaxation — that pointwise classifiers cannot

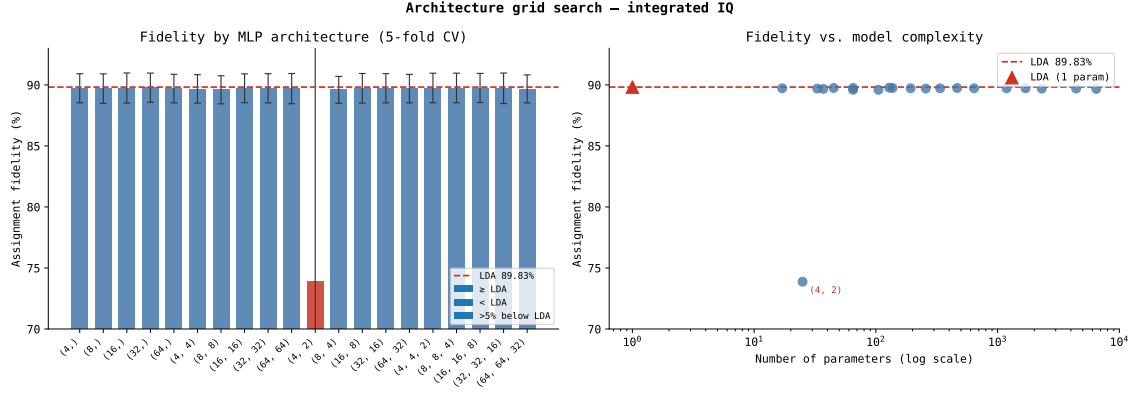


Figure 4: Architecture grid search results on integrated IQ data (5-fold CV). *Left*: assignment fidelity per MLP architecture; the red dashed line is the LDA baseline (89.83%). All architectures except the degenerate (4, 2) configuration lie within CV noise of LDA. *Right*: fidelity vs. parameter count on a log scale. No trend with model size is visible across three orders of magnitude. The (4, 2) outlier at $\approx 74\%$ reflects under-parameterisation rather than overfitting, and is excluded from Table 6.

exploit.

7.6.2 Synthetic Trace Model

Each shot is modelled as a noisy cavity response:

$$\begin{pmatrix} i(t) \\ q(t) \end{pmatrix} = \frac{e(t)}{\bar{e}} \boldsymbol{\mu}_{s(t)} + \boldsymbol{\eta}(t), \quad e(t) = 1 - e^{-t/\kappa}, \quad (13)$$

where $\bar{e} = \langle e(t) \rangle_t$ normalises the envelope so that $\langle (i(t), q(t)) \rangle_t = \boldsymbol{\mu}_{s(t)}$, matching the real blob centroids exactly. The noise $\boldsymbol{\eta}(t) \sim \mathcal{N}(\mathbf{0}, T\Sigma_s)$ is chosen so that the sample mean has covariance Σ_s , matching the real blob widths.

For $|1\rangle$ shots, qubit relaxation is modelled by drawing a relaxation time t^* from a geometric distribution with rate $1 - e^{-1/T_1}$. After t^* the signal switches from $\boldsymbol{\mu}_1$ to $\boldsymbol{\mu}_0$:

$$\boldsymbol{\mu}_{s(t)} = \begin{cases} \boldsymbol{\mu}_1 & t < t^* \\ \boldsymbol{\mu}_0 & t \geq t^* \end{cases}. \quad (14)$$

Simulation calibration. The free parameter T_1 in the simulation was chosen so that the synthetic LDA $|1\rangle$ accuracy matches the measured value of 87.3% from Table 5. This anchors the simulation to an observable that was actually measured, rather than to an assumed physical T_1 . Sweeping T_1 over the range 600–6000 samples shows that $T_1 = 3000$ samples ($\approx 15\mu\text{s}$ at 200 MSPs) reproduces the target accuracy, corresponding to a relaxation fraction of approximately 6–7% among $|1\rangle$ shots. The simulation parameters are summarised in Table 7.

A self-consistency check confirmed that rectangular integration of the synthetic traces reproduces the real `SSR0.h5` blob statistics to within 1% (Table 8).

7.6.3 CNN Architecture and Training

The `ReadoutCNN` model operates on raw traces of shape $(T, 2)$. Two `Conv1d` layers compress the time axis, global average pooling collapses it, and a small dense head produces

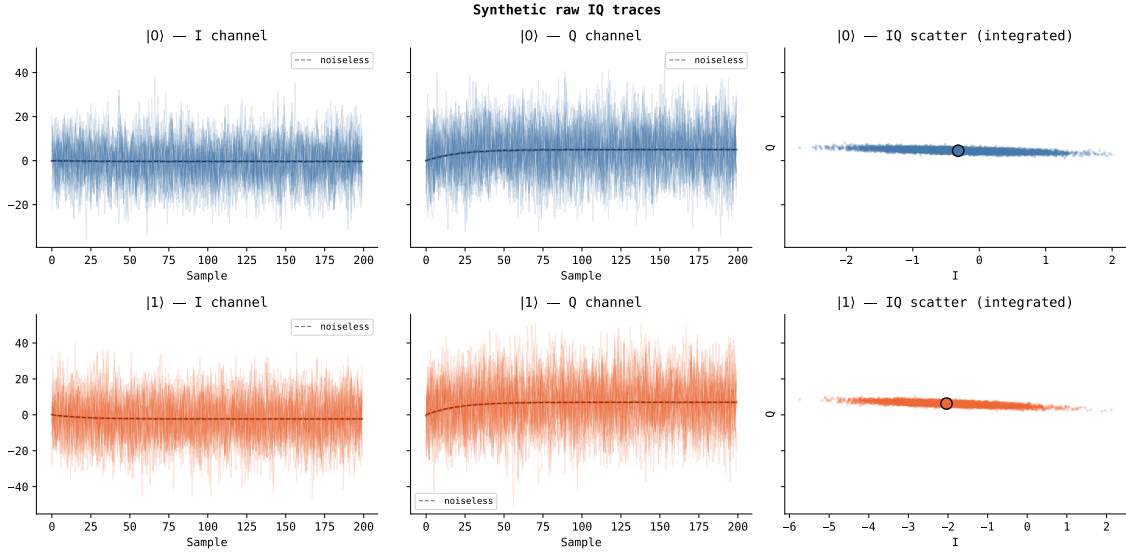


Figure 5: Synthetic raw IQ traces from the `SSR0_raw_synthetic.h5` dataset. Each panel overlays 200 noisy shots (faint) with the noiseless cavity-ringup envelope (dashed). *Top row*: $|0\rangle$ shots; both quadratures ring up and plateau near zero (the $|0\rangle$ centroid in the rotated frame). *Bottom row*: $|1\rangle$ shots; the Q channel plateaus at the $|1\rangle$ centroid. *Right column*: integrated IQ scatter after rectangular averaging; the $|1\rangle$ cloud is broader, consistent with the covariance asymmetry in Table 4. Note that the synthetic traces are generated in an unrotated frame, so the IQ axes differ from Figure 2.

Table 7: Synthetic trace simulation parameters, calibrated to match the real `SSR0.h5` LDA $|1\rangle$ accuracy of 87.3%.

Parameter	Physical meaning	Value
T	Readout window	200 samples ($\approx 1 \mu\text{s}$)
$\tau_{\text{cav}} = \kappa^{-1}$	Cavity ringup time constant	20 samples ($\approx 100 \text{ ns}$)
T_1	Qubit relaxation time	3000 samples ($\approx 15 \mu\text{s}$)
Relaxation fraction	$ 1\rangle$ shots with $t^* < T$	$\approx 6\text{--}7\%$

the binary logit for $P(|1\rangle)$. The model has 953 parameters, sized for deployment via `hls4ml` [15].

The model was trained for 100 epochs using binary cross-entropy loss and the Adam optimiser on a dataset of 10 000 shots per state.

7.6.4 Results

Table 10 compares LDA on integrated IQ against the CNN on raw traces, evaluated on the same held-out test set (20%, 4000 shots).

The CNN achieves perfect accuracy on the 91 relaxation shots in the held-out test set, compared to 41.8% for LDA — a gain of 58.2 percentage points on the shots that are entirely invisible to integration-based methods. This confirms the theoretical expectation from Section 2.6: the CNN is exploiting the temporal jump structure of the raw trace, not merely interpolating between blob positions. The $P(|1\rangle)$ confidence as a function of relaxation time t^* rises monotonically: shots that relax late in the window (spending most of the trace in $|1\rangle$) are classified with near-certainty, while shots that relax early (spending most of the trace in $|0\rangle$) are correctly identified as ambiguous.

Table 8: Self-consistency check: synthetic integrated IQ vs. real SSR0.h5.

Quantity	Synthetic	Real
$\mu_I, 0\rangle$	-0.341	-0.324
$\mu_Q, 0\rangle$	+4.521	+4.502
$\sigma_I, 0\rangle$	0.645	0.649
$\mu_I, 1\rangle$ (no relax)	-2.098	-2.095
$\mu_Q, 1\rangle$ (no relax)	+6.336	+6.313
$\sigma_I, 1\rangle$ (no relax)	0.851	0.874

Table 9: ReadoutCNN architecture summary (953 parameters).

Layer	Output shape	Parameters
Conv1d(2→8, k=8, s=2, p=3)	($\lfloor T/2 \rfloor, 8$)	136
ReLU	—	—
Conv1d(8→16, k=4, s=2, p=1)	($\cdot, 16$)	528
ReLU + GlobalAvgPool	(16,)	—
Linear(16→16)	(16,)	272
ReLU + Linear(16→1)	(1,)	17

These results should be understood as simulation estimates anchored to the real device data. The calibration procedure — adjusting T_1 until the synthetic LDA $|1\rangle$ accuracy matches the measured 87.3% — gives $T_1 = 3000$ samples ($\approx 15 \mu\text{s}$ at 200 MSps), corresponding to a relaxation fraction of approximately 5–6% of $|1\rangle$ shots within the readout window (≈ 91 relaxation events in a test set of 4000 shots). The CNN accuracy of 100% on relaxation shots should be interpreted with this sample size in mind; it reflects near-certain classification on the available test shots rather than a guarantee of perfect generalisation. The qualitative conclusion is robust: a CNN operating on raw traces has access to temporal information that the rectangular integrator discards, and this structural advantage persists regardless of the exact T_1 . Retraining on real raw traces, once collected with `acquire_decimated`, is the natural next step and is discussed in Section 8.

7.7 FPGA Deployment Path

FPGA deployment of the trained classifiers was not achieved within the scope of this work. This section outlines the concrete steps required to close the loop from the trained models to real-time discrimination inside the QICK firmware.

The current readout pipeline — rectangular integrator followed by the tProcessor threshold register — already implements LDA implicitly. Formalising this as an explicit dot-product instruction (`mathi`) adds no hardware overhead. The CNN requires a separate data path: raw decimated traces from `acquire_decimated` are fed into an `hls4ml`-synthesised IP block that produces a binary decision within approximately 100 ns to 200 ns at 250 MHz, well within the tProcessor’s feedback budget (Table 3).

The full deployment sequence is:

1. **Collect real raw traces:** connect the ZCU216 to the qubit chip and acquire `SSR0_raw.h5` using `acquire_decimated` (shape $(2, N_{\text{shots}}, T, 2)$). Set `readout_length` to at least $3/\kappa$ to capture the full cavity ringdown. (Note: the file `SSR0_raw_synthetic.h5`

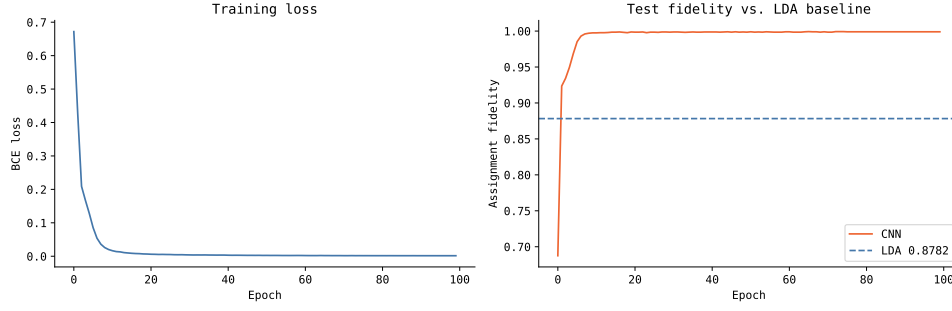


Figure 6: CNN training curves. *Left*: binary cross-entropy loss on the training set, converging smoothly within 40 epochs. *Right*: test-set assignment fidelity vs. epoch, reaching $\approx 99.9\%$ and remaining well above the LDA baseline of 0.878 (dashed). The absence of overfitting is consistent with the small model size (953 parameters) and large dataset (10 000 shots per class).

Table 10: CNN vs. LDA on synthetic raw traces. LDA operates on the time-averaged (I, Q) ; the CNN operates on the full trace $(T, 2)$. Simulation calibrated so that the synthetic LDA $|1\rangle$ accuracy matches the measured value of 87.3%. All values from the held-out test set ($n = 4000$; ≈ 91 relaxation shots).

Classifier	$\mathcal{F}$	$ 0\rangle$ acc.	$ 1\rangle$ acc. (all)	$ 1\rangle$ acc. (relax. only)
LDA (integrated IQ)	0.878	0.914	0.843	0.418
1D CNN (raw trace)	0.999	1.000	0.998	1.000
CNN gain	+12.1%	+8.6%	+15.5%	+58.2%

used in Part III is a simulated dataset, not a hardware capture; this step acquires the real counterpart.)

2. **Retrain and validate**: retrain the CNN on real traces, using the same calibration approach — adjust T_1 in the synthetic model until the simulated LDA $|1\rangle$ accuracy matches the measured device value, then use the calibrated synthetic dataset to supplement real training data.
3. **Export to HLS**: export the trained PyTorch model to ONNX, then convert using `hls4ml` [15] targeting the ZCU216 fabric (`xczu49dr-ffvf1760-2-e`), `io_stream`, and fixed-point precision `ap_fixed<16,6>`.
4. **Synthesise and integrate**: synthesise the HLS project in Vivado and integrate the resulting IP block into the QICK bitstream alongside the existing readout chain.
5. **Benchmark**: compare LDA-threshold, CNN-IP, and host-side discrimination on the same experimental data to quantify latency reduction and fidelity gain under realistic qubit conditions.

Based on the model’s 953 parameters and typical `hls4ml` resource scaling for comparable architectures, the CNN IP block is expected to require on the order of 10 000 LUTs and tens of DSP slices — a small fraction of the ZCU216’s available fabric — though this must be confirmed by synthesis.

8 Conclusions and Outlook

8.1 Summary

This thesis has documented the setup, characterisation, and initial evaluation of a QICK-based readout architecture for superconducting qubits on a Xilinx ZCU216 RFSoc platform at Qilimanjaro Quantum Tech, and has explored the design of a machine-learning-assisted state discrimination pipeline using existing experimental data.

The main contributions are:

1. **System integration:** The ZCU216 board, XM655 analogue front-end, and QICK firmware were brought up from scratch, including SSH access, PYNQ configuration, and a persistent Pyro4 remote-operation server that preserves calibration state across notebook sessions.
2. **Phase-coherent readout calibration:** A systematic calibration procedure was developed that measures the random DAC–ADC DDS phase offset at board power-on. The linear phase-vs-frequency model $\phi(f) = \phi_0 - 360^\circ \tau f$ was validated, and the fitted `res_phase` value was incorporated into the standard experiment configuration.
3. **Readout data-path:** The trade-offs between decimated (waveform) and accumulated (averaged IQ) readout modes were characterised. The accumulated mode produces one integrated (I, Q) pair per shot, suitable for SSRO; repeat-averaging across shots improves spectroscopy SNR but destroys single-shot information and cannot be used for state discrimination. The resonator spectroscopy workflow was established for both single-qubit and multiplexed scenarios.
4. **Real-time feedback primitives:** The tProcessor V2 conditional branch (`condj`) was demonstrated in loopback as the core primitive for mid-circuit measurement. An active qubit reset sequence was designed, combining dispersive readout, thresholding, and a calibrated π pulse, with an estimated digital decision-to-fire latency below 100 ns.
5. **ML-assisted discrimination:** A complete offline discrimination pipeline was developed and evaluated on existing experimental SSRO data. On integrated IQ data, LDA achieves an assignment fidelity of 90.6% ($|0\rangle$: 94.0%, $|1\rangle$: 87.3%), and a systematic grid search over 20 MLP architectures confirms that no nonlinear classifier improves on this — a consequence of the near-Gaussian, linearly separable IQ blob structure. The $|1\rangle$ accuracy deficit is identified as the signature of T_1 relaxation mid-shot, which integration-based methods cannot recover. A lightweight 1D CNN (953 parameters) trained on physically-grounded synthetic raw traces achieves 99.9% overall fidelity and 100% accuracy on the 91 relaxation shots in the test set, compared to 41.8% for LDA on the same shots — a gain of 58.2 percentage points. The 100% figure reflects near-certain classification on the available test shots and should be interpreted in light of the small relaxation sample size. FPGA deployment of the CNN was not achieved within the scope of this work; Section 7.7 outlines the concrete path toward it.

8.2 Outlook

Several natural extensions of this work follow directly from the results above:

CNN retraining on real raw traces. The CNN results in Chapter 7 are based on synthetic traces anchored to real IQ statistics. The first experimental priority is to collect a real `SSRO_raw.h5` dataset from the ZCU216 using `acquire_decimated`, retrain the CNN, and validate that the T_1 -relaxation recovery generalises to real data. The synthetic T_1 parameter should be recalibrated to match the measured $|1\rangle$ error rate ($\approx 12.75\%$, implying $T_1 \approx 1200\text{--}1500$ samples rather than the 600 used in simulation) before the synthetic-to-real transfer is attempted.

On-FPGA classifier deployment. With real raw traces in hand, the CNN can be exported via `hls4ml` and synthesised as a fixed-point IP block in the QICK bitstream, as described in Section 7.7. This closes the active-reset feedback loop at full speed: the tProcessor receives a binary decision from the CNN IP block within 100 ns–200 ns, without a software round-trip.

Multi-qubit readout. The current architecture supports a single readout tone. Extending to multiplexed readout via the QICK PFB firmware would allow simultaneous readout of all qubits in Qilimanjaro’s multi-qubit chips, which is a prerequisite for running quantum error-correction cycles.

Integration with Qilimanjaro’s software stack. The QICK-based readout layer developed here should be integrated with Qilimanjaro’s higher-level quantum programming interfaces (circuit compilation, calibration databases, experiment management) to provide a production-ready control stack.

Fluxonium-specific readout optimisation. The dispersive shift landscape for fluxonium qubits is two-dimensional in the (Φ_z, Φ_x) flux space. An ML-assisted optimisation of the readout circuit parameters — coupling capacitance, resonator frequency — to maximise the dimensionless cavity pull over a large flux region is a complementary research direction being pursued in a related BSc project within the team.

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A Key Code Listings

This appendix collects the most important Python/QICK code snippets used in this thesis. Full notebooks are available in the accompanying GitHub repository.

A.1 Phase Calibration Programme

```

1  class PhaseCalProgram(AveragerProgram):
2      """Sweeps frequency and measures DAC-ADC phase offset."""
3
4      def initialize(self):
5          cfg = self.cfg
6          self.declare_gen(ch=cfg["res_ch"], nqz=1)
7          self.declare_readout(ch=cfg["ro_ch"],
8                               length=cfg["readout_length"],
9                               freq=cfg["pulse_freq"],
10                              gen_ch=cfg["res_ch"])
11          freq_reg = self.freq2reg(cfg["pulse_freq"],
12                                   gen_ch=cfg["res_ch"],
13                                   ro_ch=cfg["ro_ch"])
14          self.set_pulse_registers(ch=cfg["res_ch"],
15                                   style="const",
16                                   freq=freq_reg,
17                                   phase=0,
18                                   gain=cfg["pulse_gain"],
19                                   length=cfg["length"])
20
21          self.synci(200)
22
23      def body(self):
24          self.measure(pulse_ch=self.cfg["res_ch"],
25                      adcs=[self.cfg["ro_ch"]],
26                      adc_trig_offset=self.cfg["adc_trig_offset"],
27                      wait=True,
28                      syncdelay=self.us2cycles(self.cfg["relax_delay"]))

```

A.2 Fast Amplitude Sweep with RAveragerProgram

```

1  class SweepProgram(RAveragerProgram):
2      """Sweeps DAC gain inside the FPGA without Python overhead."""
3
4      def initialize(self):
5          cfg = self.cfg
6          self.declare_gen(ch=cfg["res_ch"], nqz=1)
7          self.declare_readout(ch=cfg["ro_ch"],
8                               length=cfg["readout_length"],
9                               freq=cfg["pulse_freq"],
10                              gen_ch=cfg["res_ch"])
11          freq_reg = self.freq2reg(cfg["pulse_freq"],
12                                   gen_ch=cfg["res_ch"],
13                                   ro_ch=cfg["ro_ch"])
14          self.set_pulse_registers(ch=cfg["res_ch"],
15                                   style="const",
16                                   freq=freq_reg,
17                                   phase=cfg["res_phase"],
18                                   gain=cfg["start"],
19                                   length=cfg["length"])

```

```

20     self.r_rp    = self.ch_page(cfg["res_ch"])
21     self.r_gain = self.sreg(cfg["res_ch"], "gain")
22     self.synci(200)
23
24     def body(self):
25         self.measure(pulse_ch=self.cfg["res_ch"],
26                     adcs=[self.cfg["ro_ch"]],
27                     adc_trig_offset=self.cfg["adc_trig_offset"],
28                     wait=True,
29                     syncdelay=self.us2cycles(self.cfg["relax_delay"]))
30
31     def update(self):
32         self.mathi(self.r_rp, self.r_gain, self.r_gain,
33                  '+', self.cfg["step"])

```

A.3 Active Reset Sequence (Pseudocode)

Algorithm 1 Active qubit reset

Require: Calibrated π -pulse amplitude A_π and readout threshold θ

Trigger ADC readout window

Play readout tone on resonator channel

wait for readout window to close

$b \leftarrow \text{FPGA_threshold}(I + jQ, \theta)$

$\triangleright b = 1$ if qubit in $|1\rangle$

regwi $r_b \leftarrow b$

condj $r_b < 1$ **goto** SKIP_PI

Play π pulse on qubit drive channel

label SKIP_PI

synci t_{relax}

$\triangleright$ Allow cavity photons to decay
